

# M.I.N.D

Mining Interesting Narratives in Discussion

20 Research Questions on Student Discussion Data

5 Discussions · 294 Responses · 69 Students

**D1: Motivation**

**D2: Preprocessing**

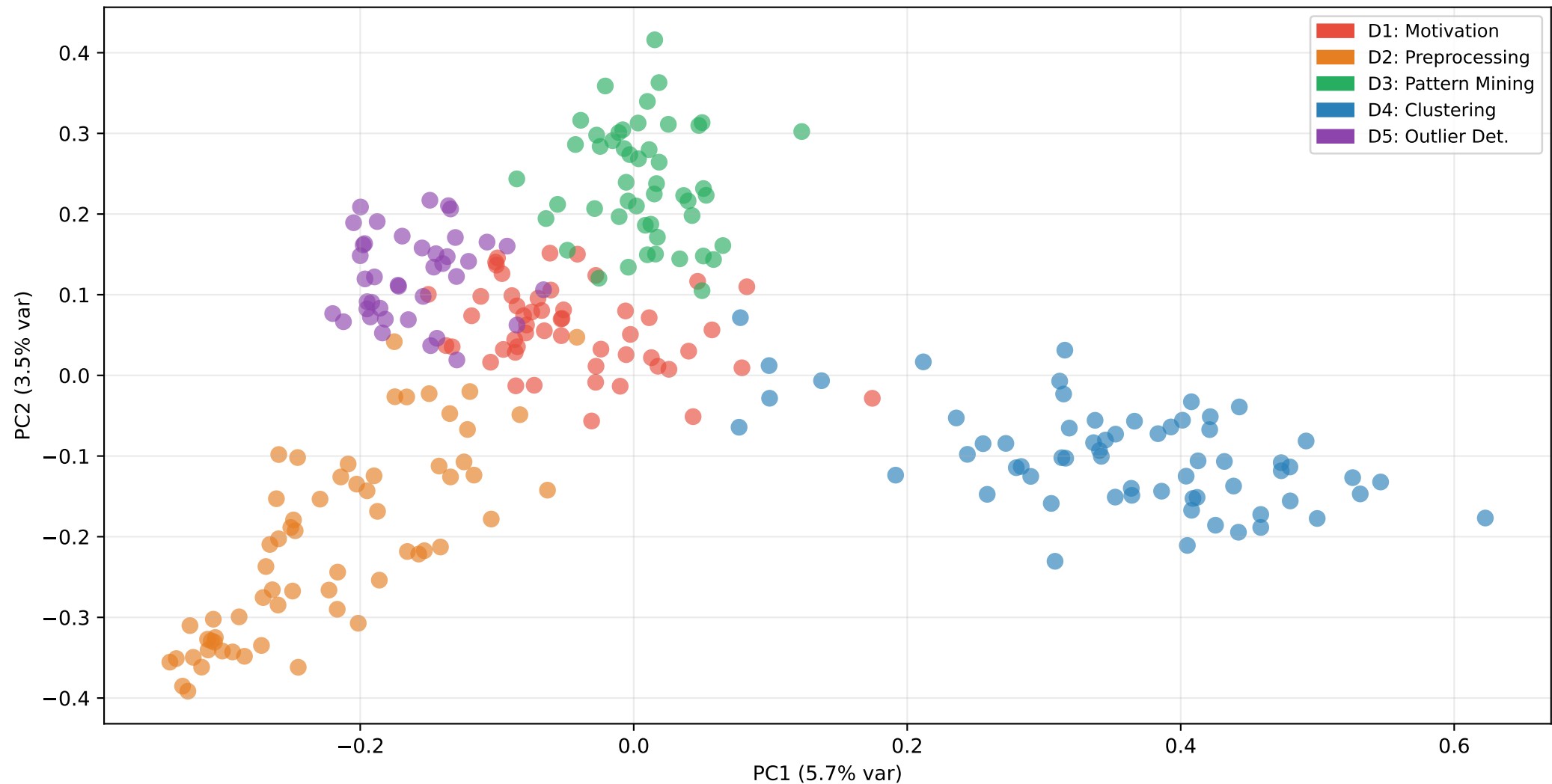
**D3: Pattern Mining**

**D4: Clustering**

**D5: Outlier Det.**

Pattern Mining · Clustering · Outlier Detection

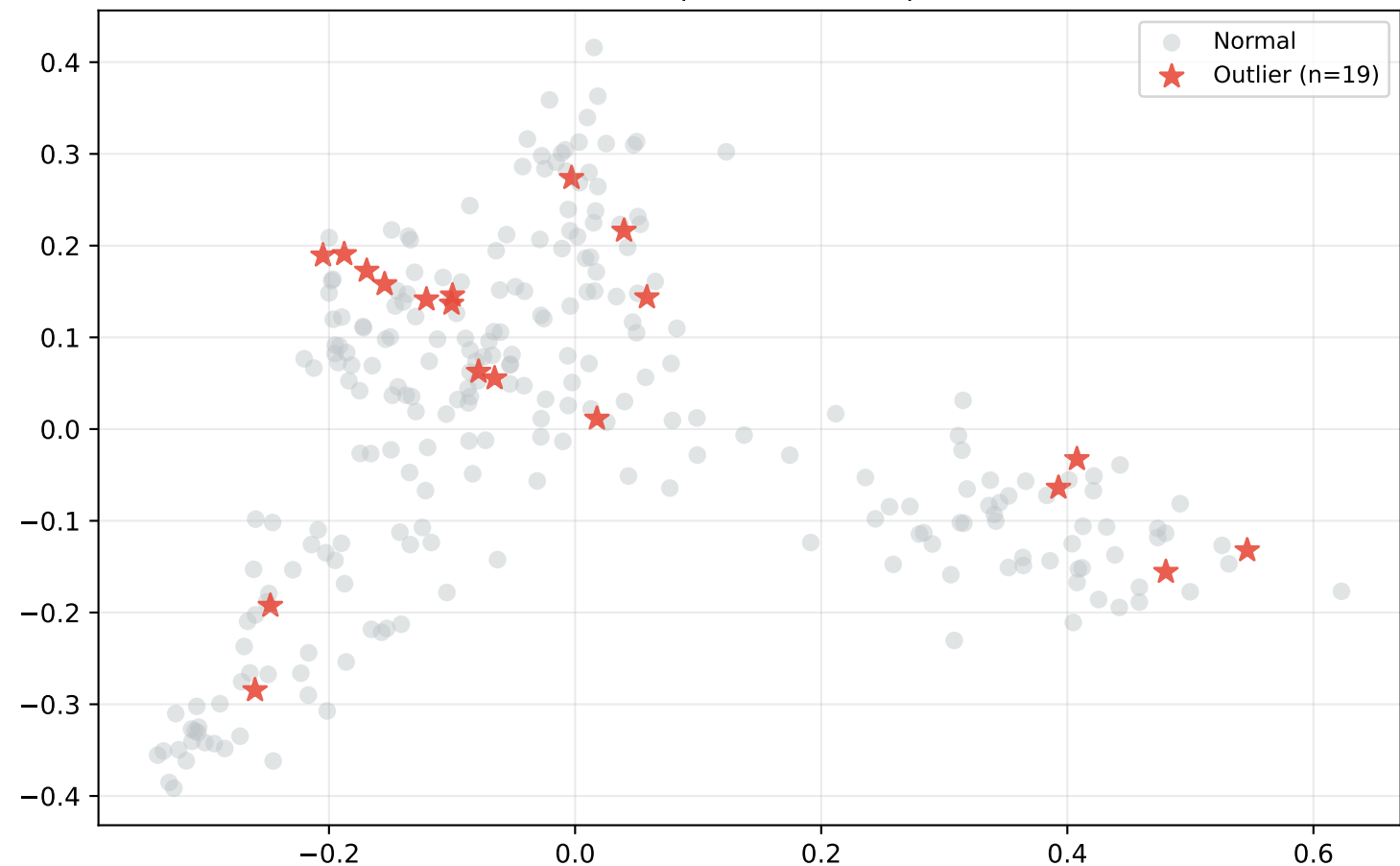
## Q01 | What Is the 2D Semantic Landscape of All Student Responses?



Key Finding: TF-IDF + PCA explains 9.2% variance. Discussions form loose but visible regional clusters — D1 and D5 occupy opposite ends, reflecting maximally different topics.

Q02 | Which Responses Are Semantic Outliers Across the Entire Corpus?

Outlier Responses in PCA Space

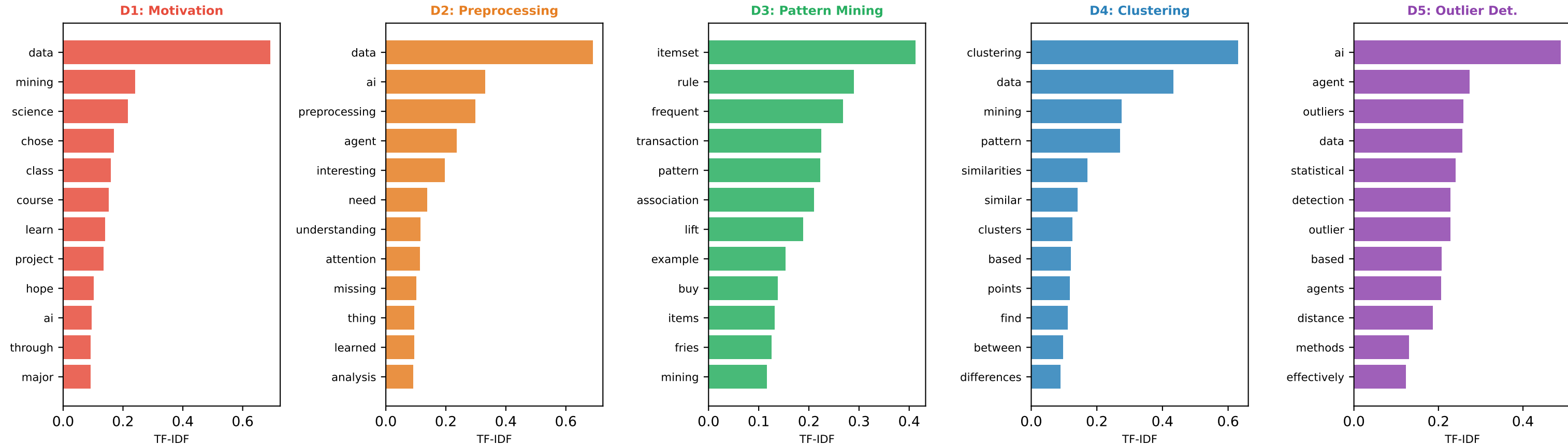


Detected Outlier Responses

| Disc | Response Preview                                  |
|------|---------------------------------------------------|
| D1   | The course would teach me how to connect dots     |
| D1   | I was interested in methods that can extract inte |
| D1   | I'm interested in AI topics such as machine learn |
| D1   | I want to learn the fundamental concepts of data  |
| D1   | Biggest reason: class is on Friday 9AM - needed   |
| D2   | The most interesting thing I learned today has to |
| D2   | The most interesting thing I learned today is kno |
| D3   | natural language processing. Every conversation   |

Key Finding: Isolation Forest (7% contamination) flagged 19 outliers. Outliers span multiple discussions — unusual thinking is not confined to one topic.

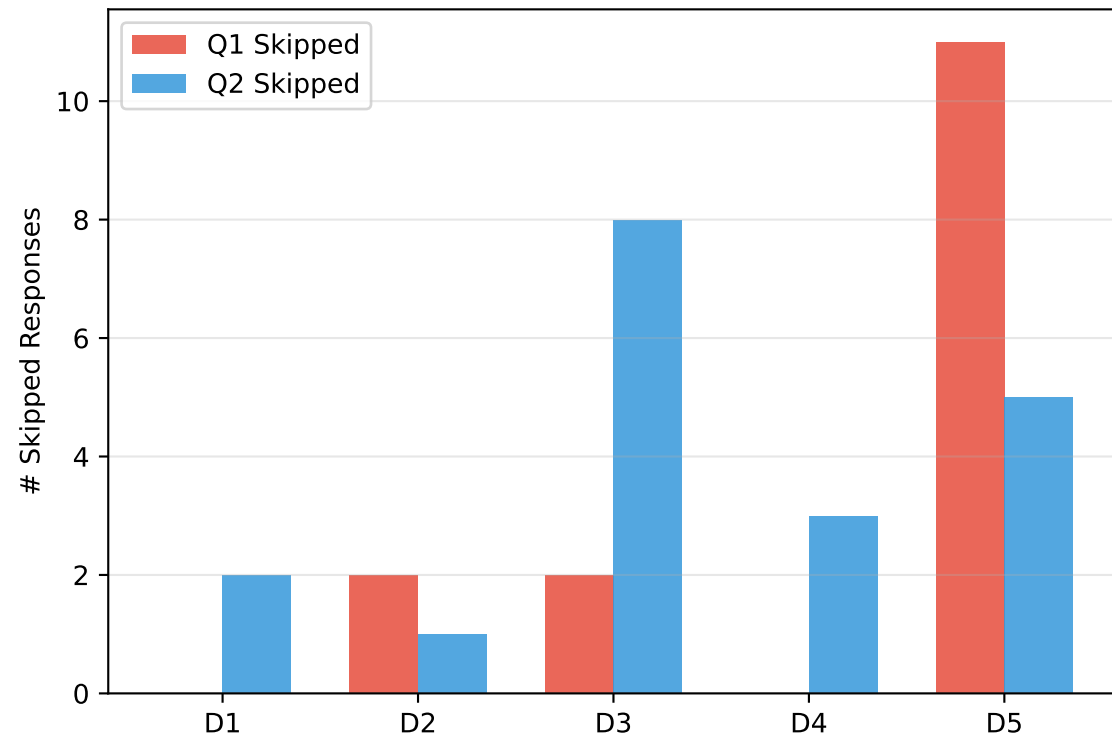
### Q03 | What Words Define Each Discussion Uniquely? (Discussion DNA)



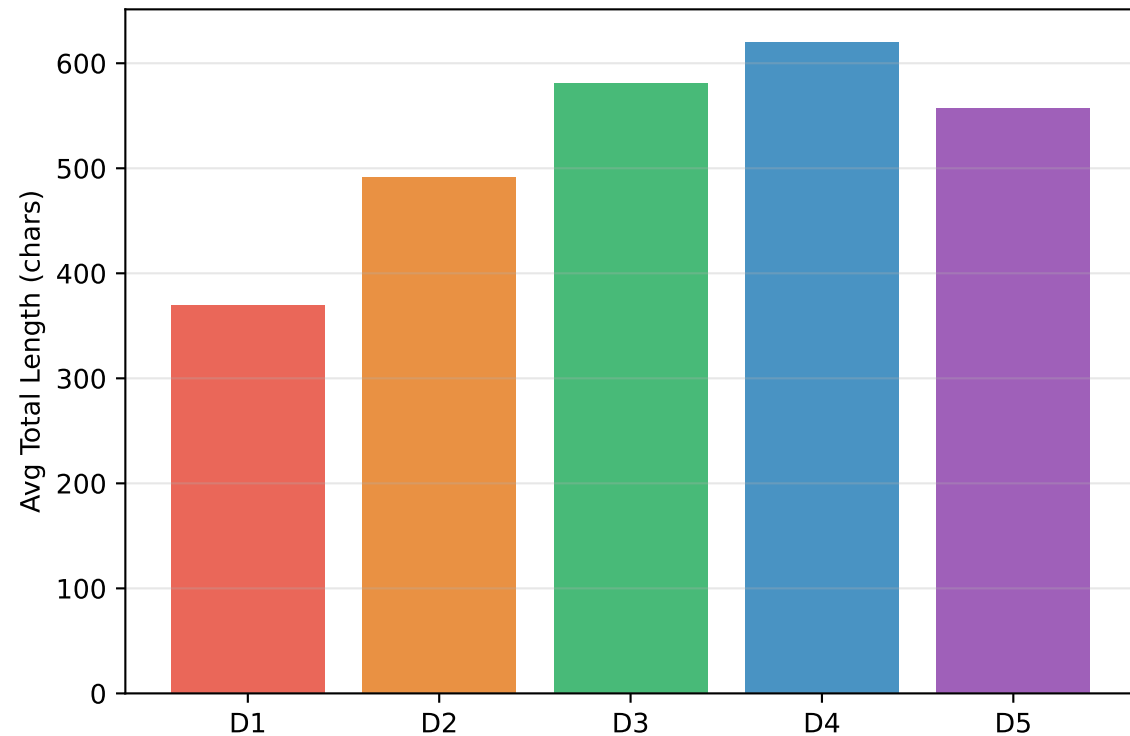
Key Finding: Each discussion has a distinct vocabulary fingerprint visible in TF-IDF. D1: "learn/chose/course". D5: "outlier/detection/statistical". The curriculum is in the words.

## Q04 | Which Discussion Had Students Giving Up? (Null & Length Analysis)

### Skipped Responses by Discussion



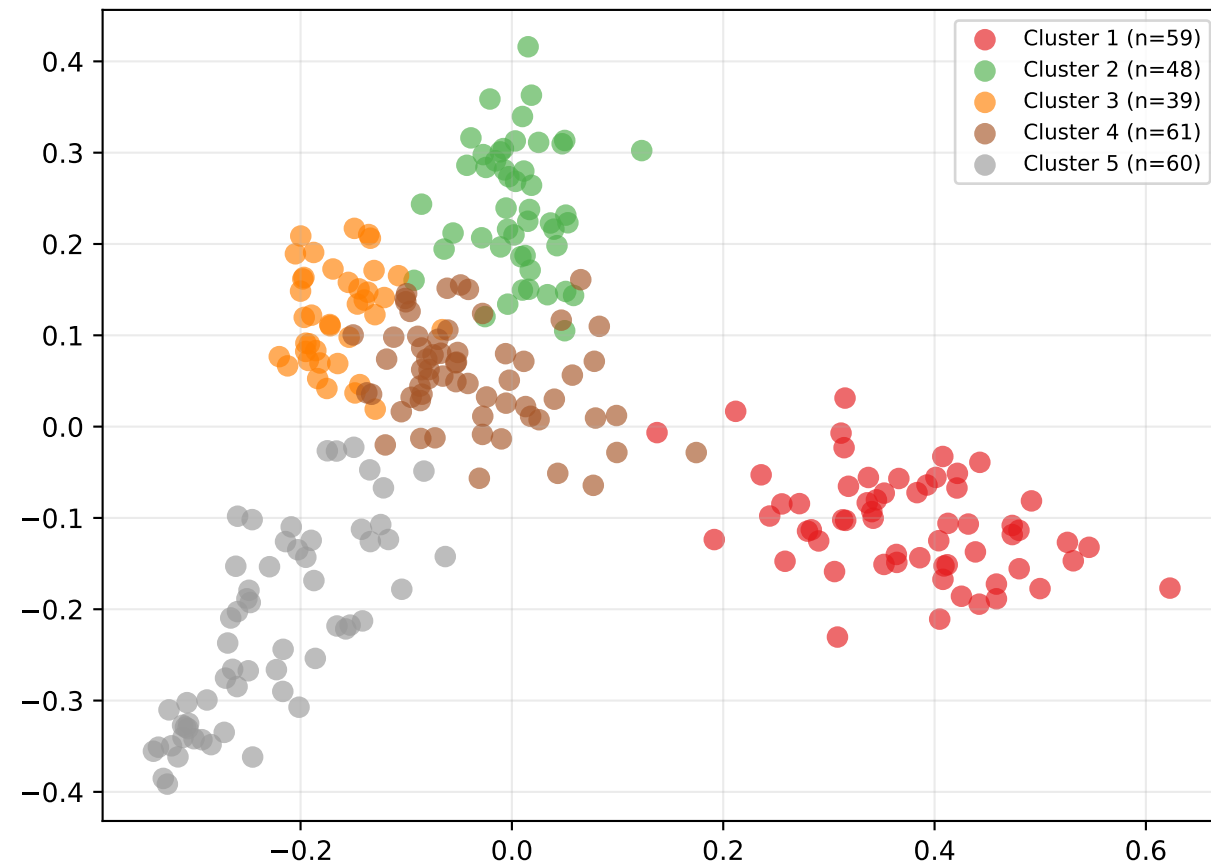
### Avg Response Length by Discussion



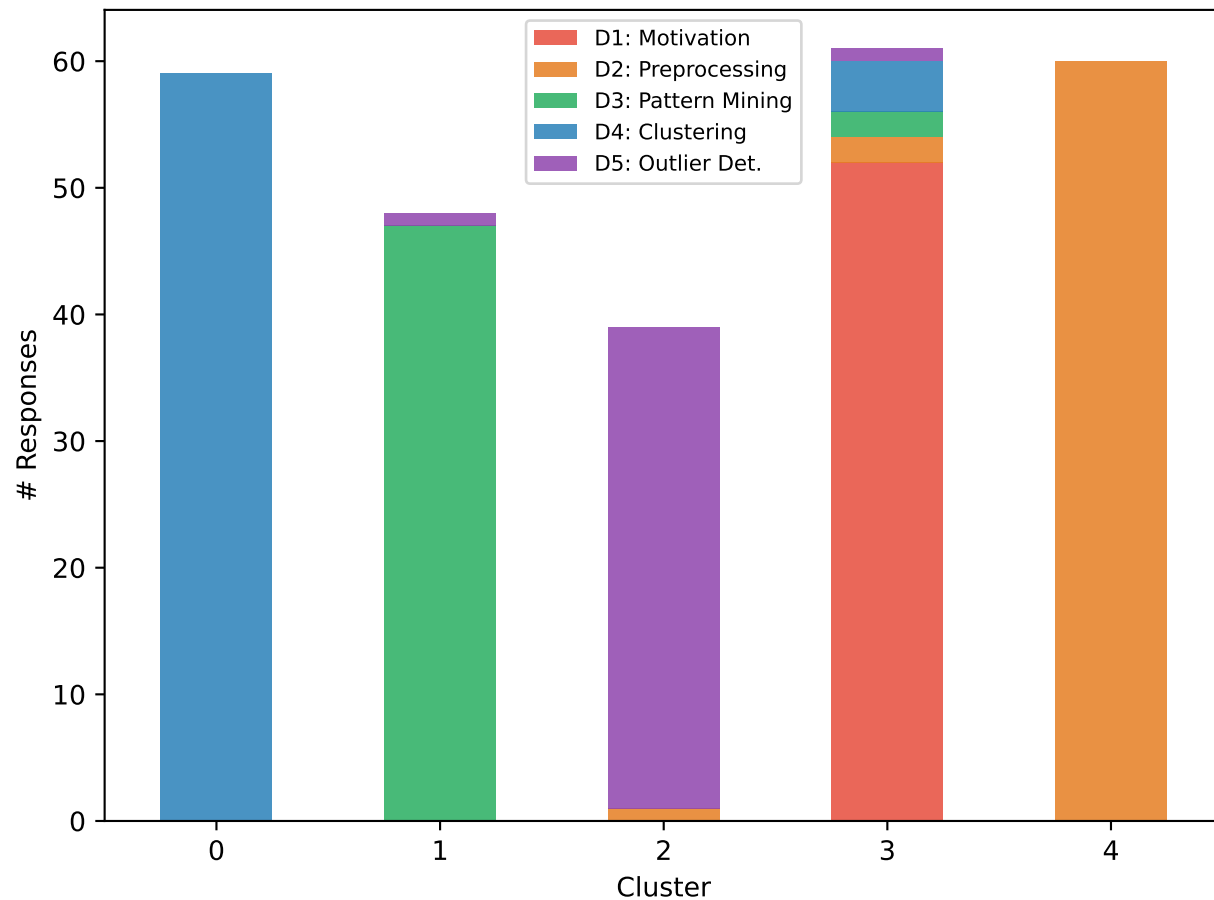
Key Finding: D5 had the most skipped responses. D5-Q1 alone had 11 nulls. Average response length peaks at D4 — clustering inspired the longest answers.

## Q05 | Are There Natural Topic Tribes Across All Responses?

K-Means Clusters (k=5) — Ignoring Discussion Labels

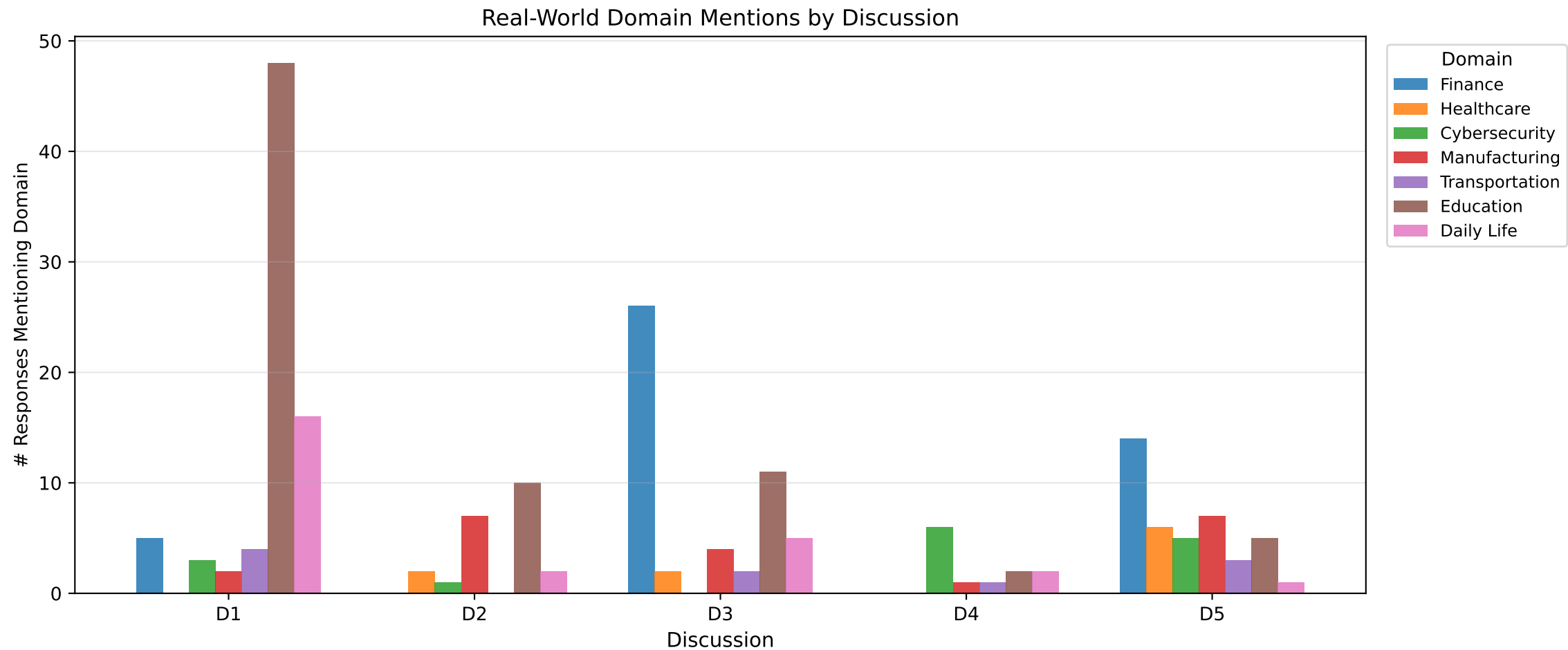


Cluster Composition by Discussion



Key Finding: K-Means (k=5) shows clusters do NOT map cleanly to discussions — every cluster contains responses from multiple discussions. Thinking styles transcend topic.

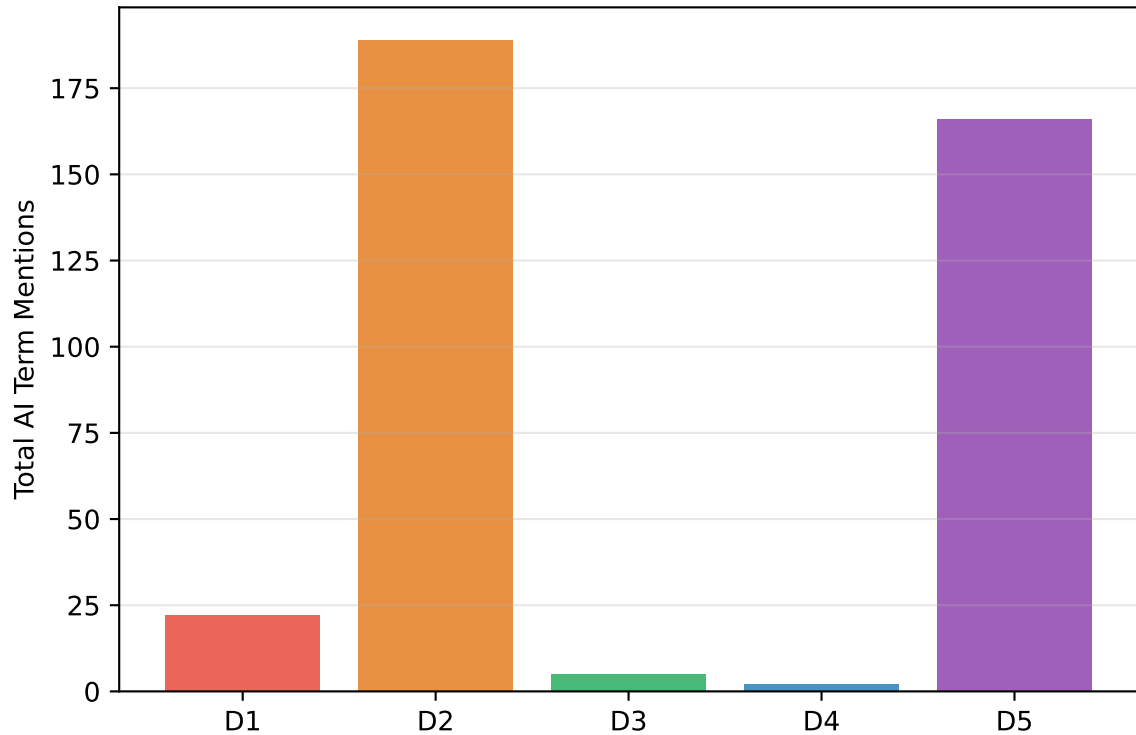
Q06 | What Real-World Domains Do Students Default To?



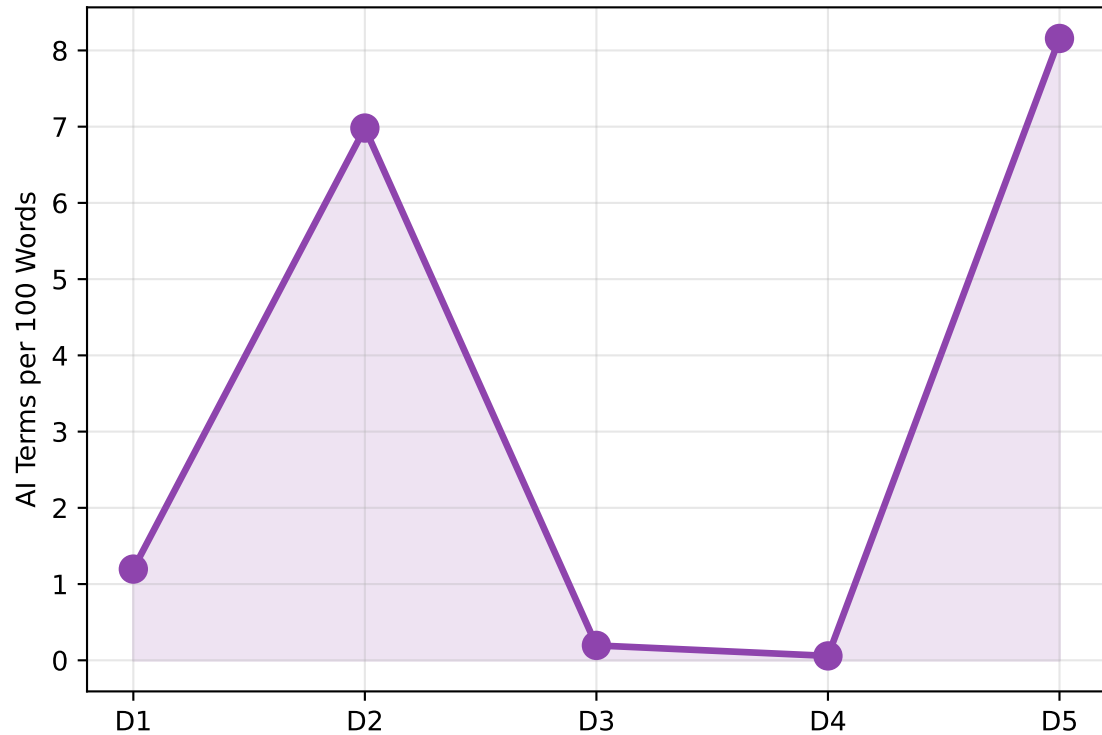
Key Finding: "Education" is the most cited domain overall. Finance dominates D5. Daily Life peaks in D1. Students reach for finance examples regardless of topic.

## Q07 | How Did AI-Related Language Appear Across All Discussions?

Raw AI Term Count per Discussion



AI Term Density per Discussion

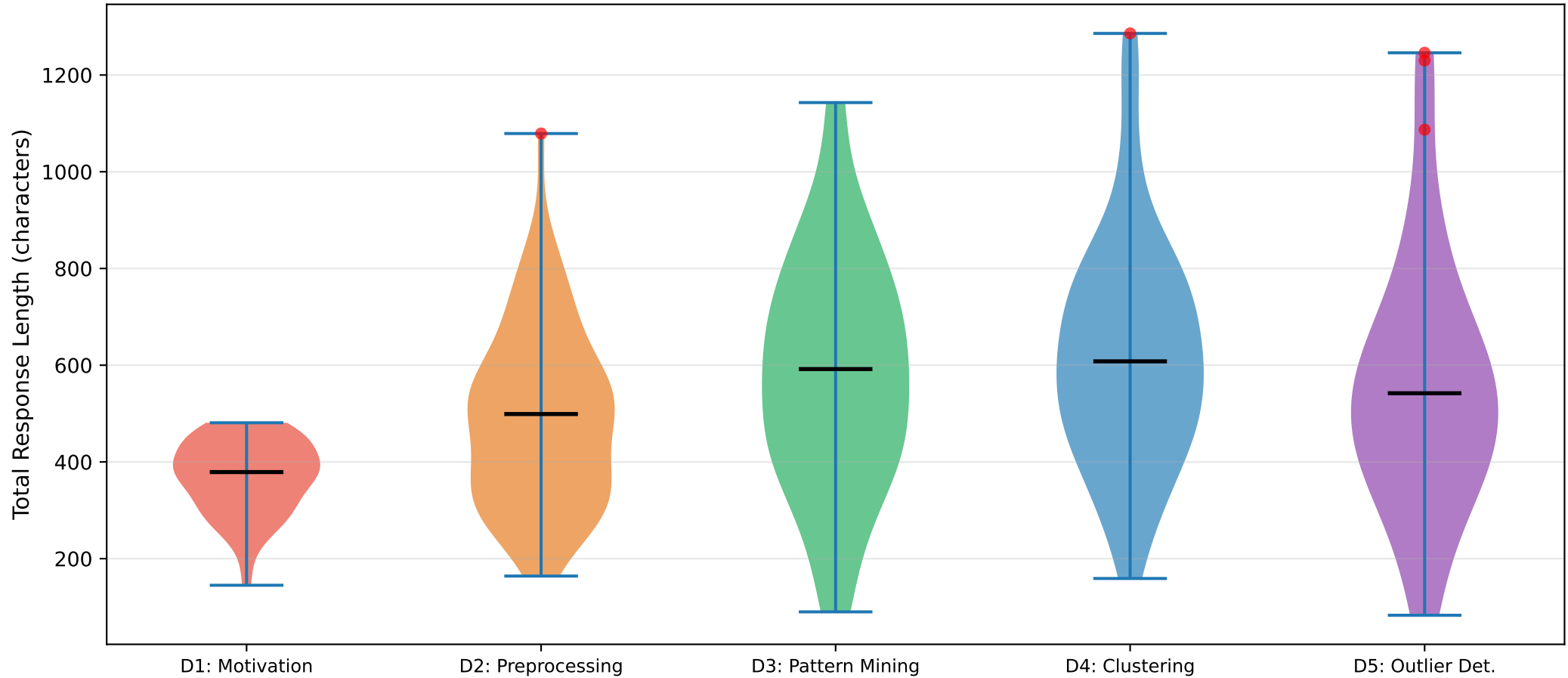


Key Finding: AI language peaks at D5. D1-D3 show low AI density — students were not thinking in AI terms when discussing motivation or pattern mining.



## Q08 | Who Wrote the Most and the Least?

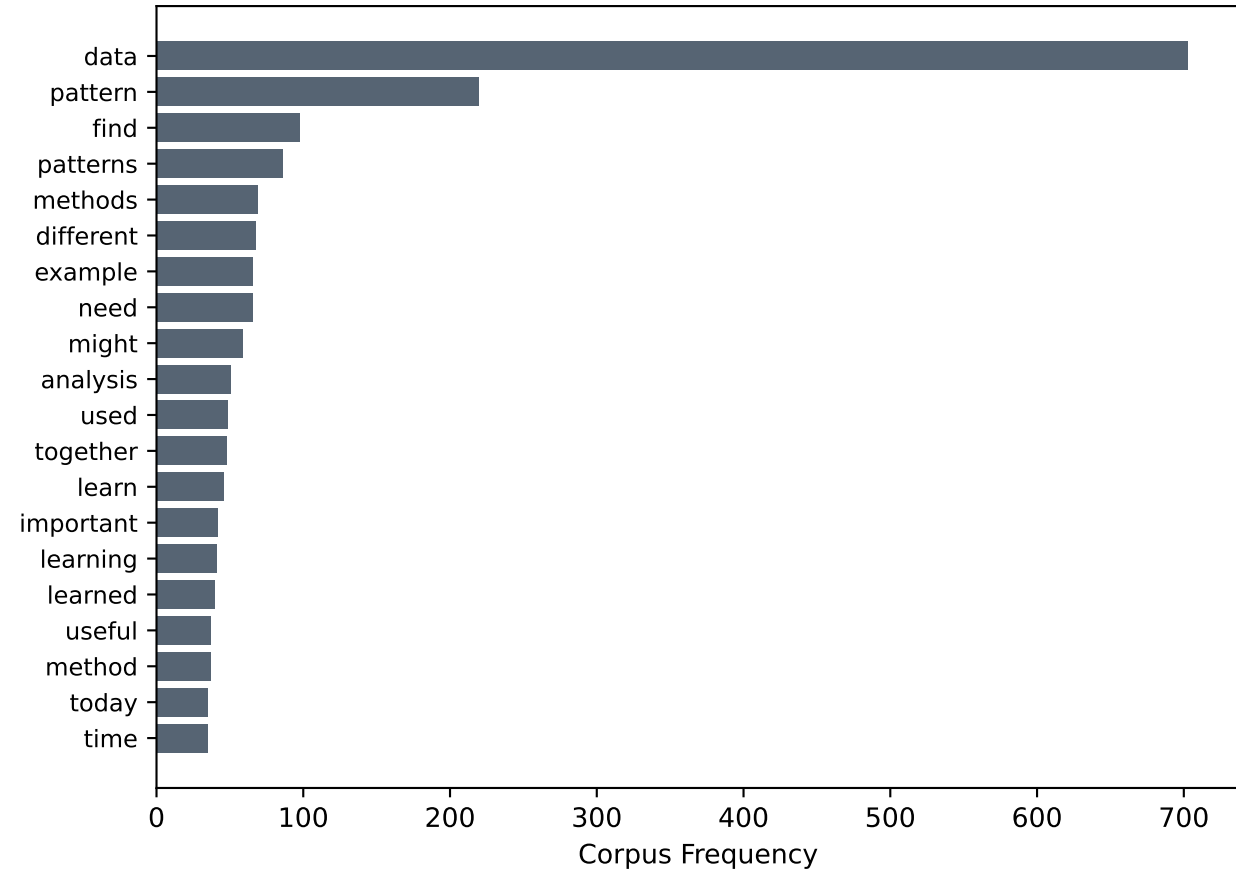
Response Length Distribution by Discussion



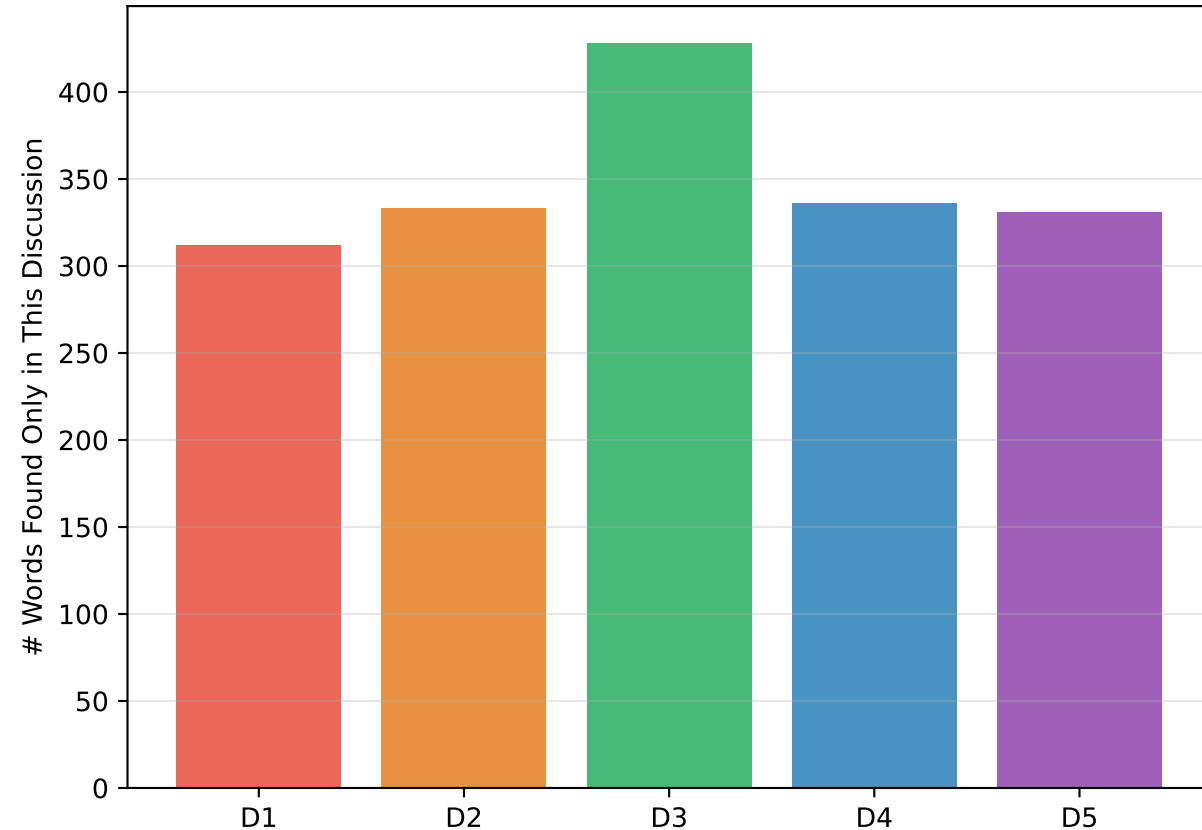
Key Finding: D5 has the widest spread in response length — highest inequality in effort. Red dots = statistical outliers ( $1.5 \times \text{IQR}$ ). Some students wrote 1200+ chars; others barely 80.

## Q09 | What Words Appear in ALL 5 Discussions vs. Only One?

Top Universal Words  
(in all 5 discussions — 59 total)



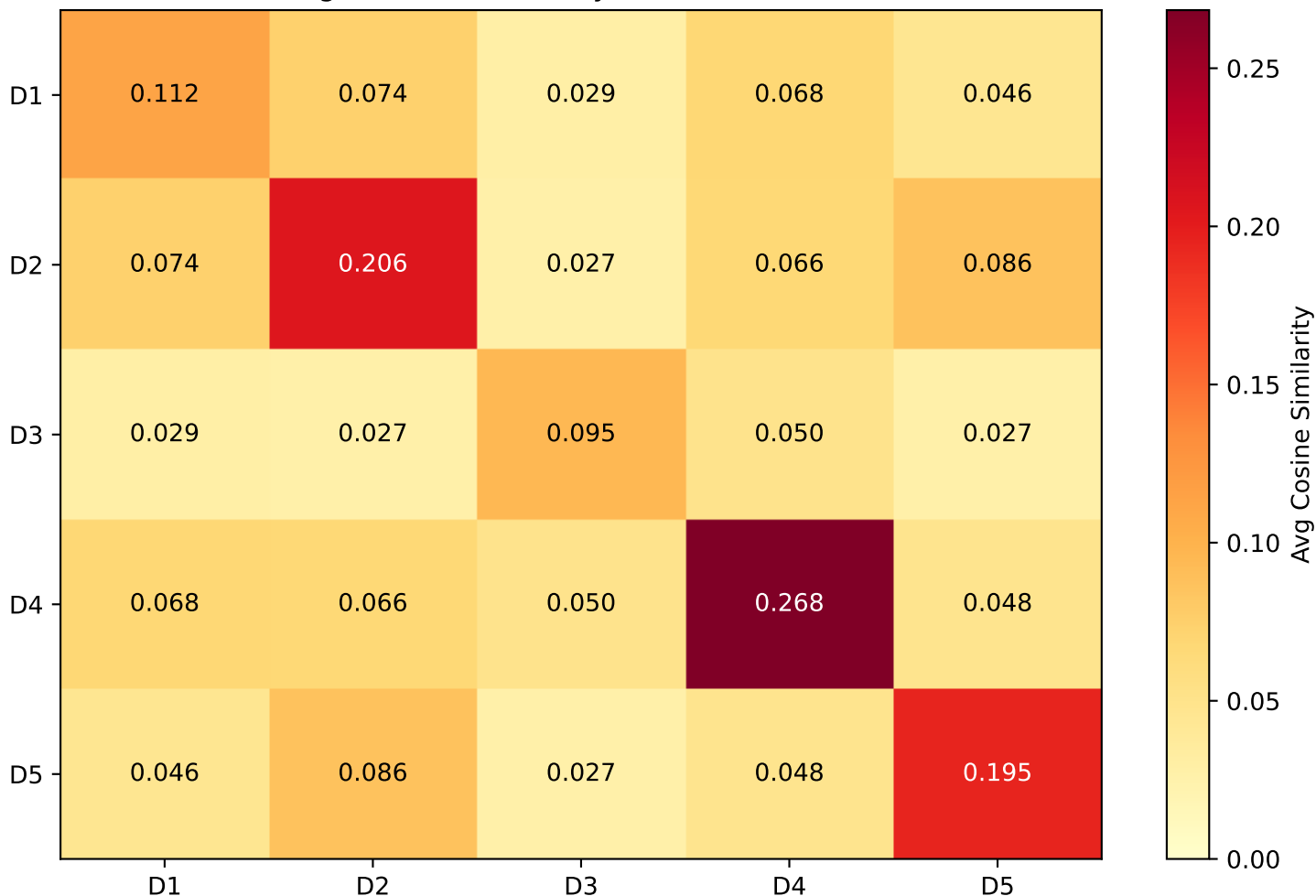
Exclusive Vocabulary per Discussion



Key Finding: 59 words are universal (appear in every discussion). D3 has the most exclusive vocabulary — its topic brings the most unique language.

# Q10 | Are Responses Within the Same Discussion More Similar to Each Other?

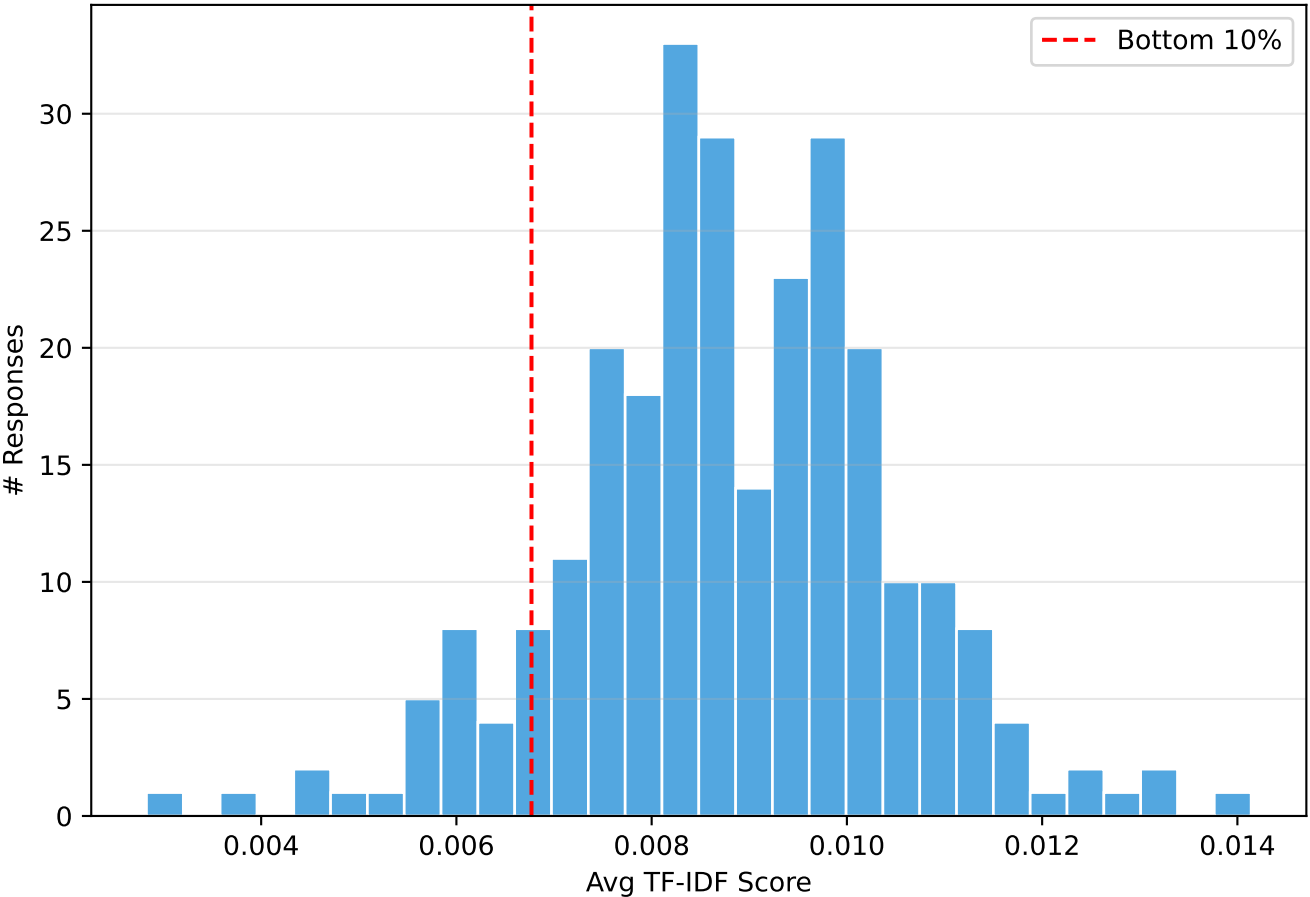
Average Cosine Similarity Between Discussions



Key Finding: Within-discussion avg similarity: 0.175 vs cross-discussion: 0.052. D4-D5 share the highest cross-similarity — clustering and outlier detection vocabularies overlap.

Q11 | Which Responses Are the Most Generic and Formulaic?

Distribution of Response Uniqueness



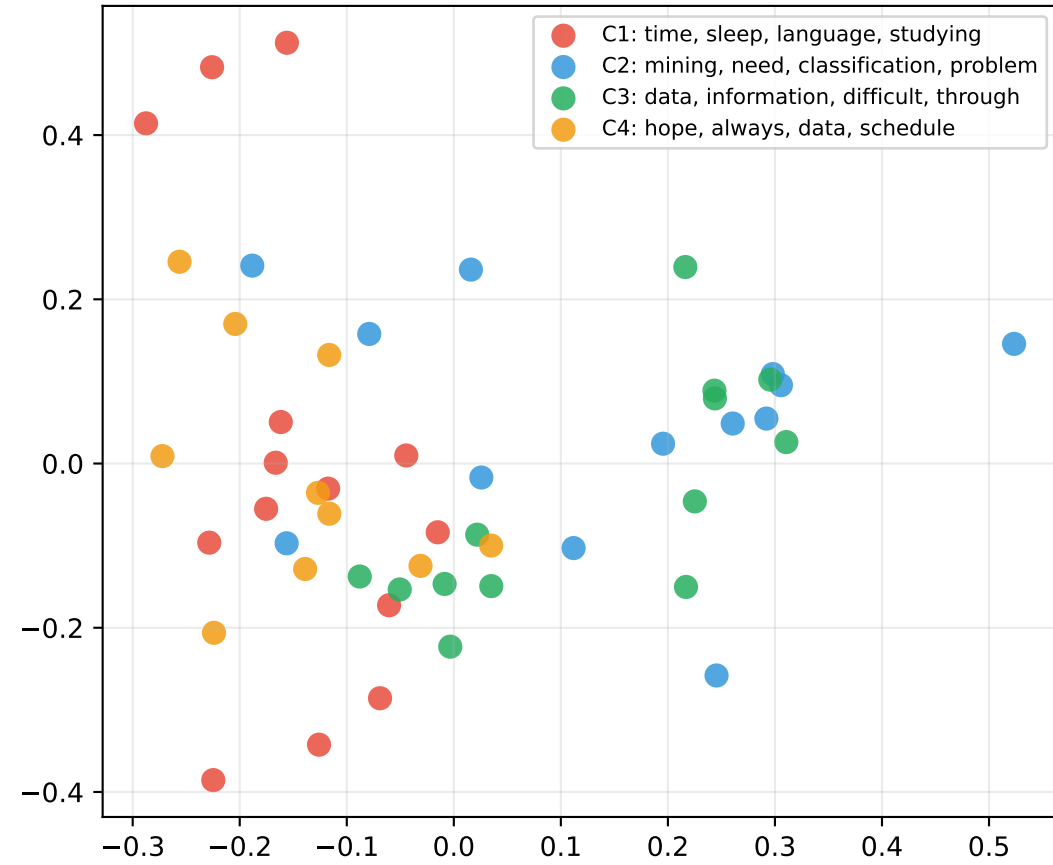
Most Generic Responses

| D  | Response Preview                                                  | Score   |
|----|-------------------------------------------------------------------|---------|
| D3 | YouTube video recommendation                                      |         |
| D3 | UX aspect.                                                        | 0.00283 |
| D3 | A supermarket analyzes millions of                                | 0.00392 |
| D5 | A. detecting unusual amount of mo                                 | 0.00441 |
| D1 | I wanted to place all classes in one                              | 0.00467 |
| D5 | We can react to server disaster qui                               | 0.00508 |
| D3 | We can use LIFT criteria.<br>LIFT solves by mitigating the correl | 0.00540 |
| D2 | Society. The people relation the mo                               | 0.00555 |
| D3 | It is a restaurant, for example, Mea                              | 0.00560 |
| D3 | Netflix recommendation.<br>Transaction: { Avengers, Spiderman     | 0.00571 |
| D1 | I took this course because I had no                               | 0.00573 |

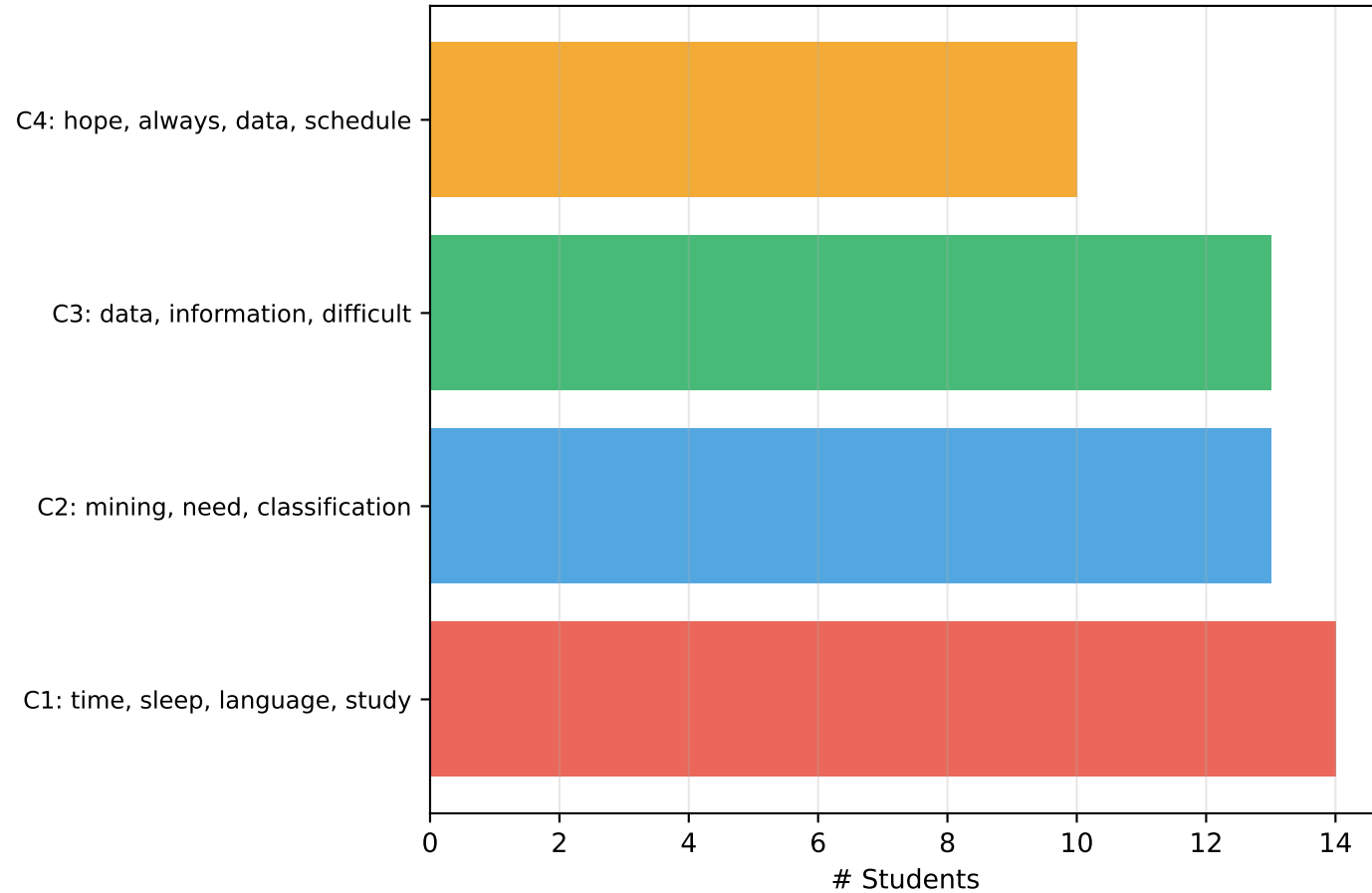
Key Finding: Responses with the lowest avg TF-IDF use only common, non-distinctive words. These are textbook answers — correct but adding no unique perspective.

## Q12 | What Types of Life Problems Did Students Want to Solve? (D1-Q2)

D1-Q2 Personal Problems: 4 Clusters



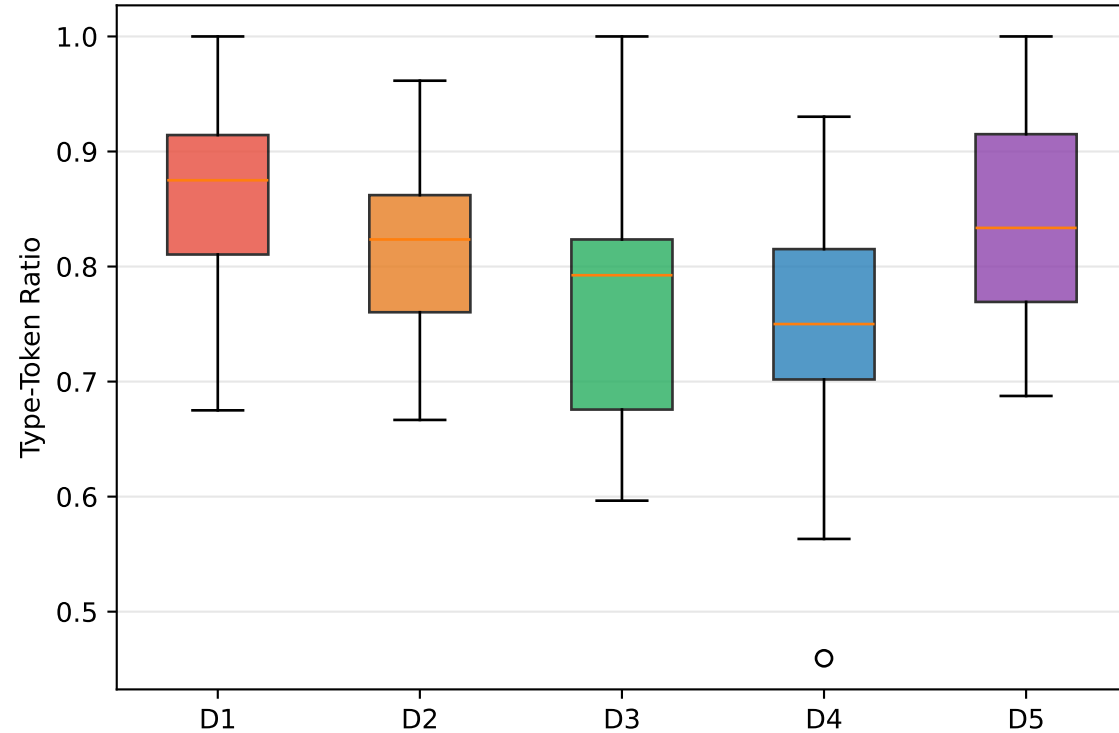
Students per Problem Cluster



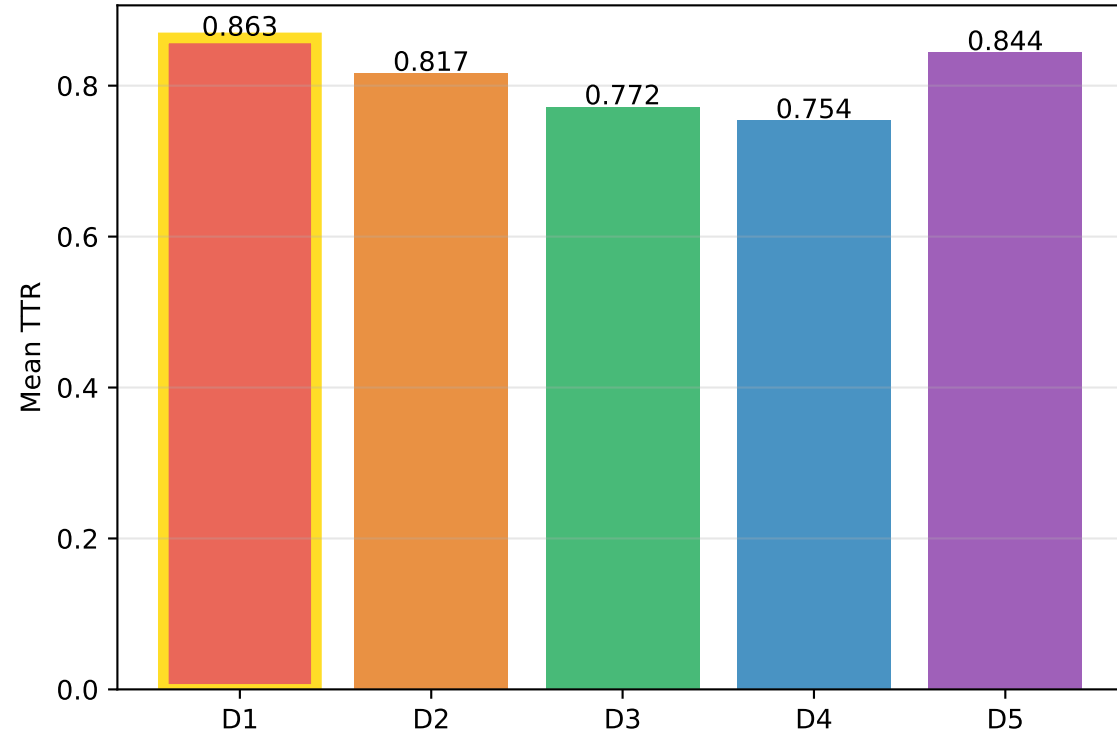
Key Finding: K-Means ( $k=4$ ) on D1-Q2 reveals problem archetypes: productivity/scheduling, finance/spending, transportation/commuting, and data organization projects.

## Q13 | Which Discussion Had the Richest, Most Diverse Vocabulary?

Vocabulary Richness Distribution



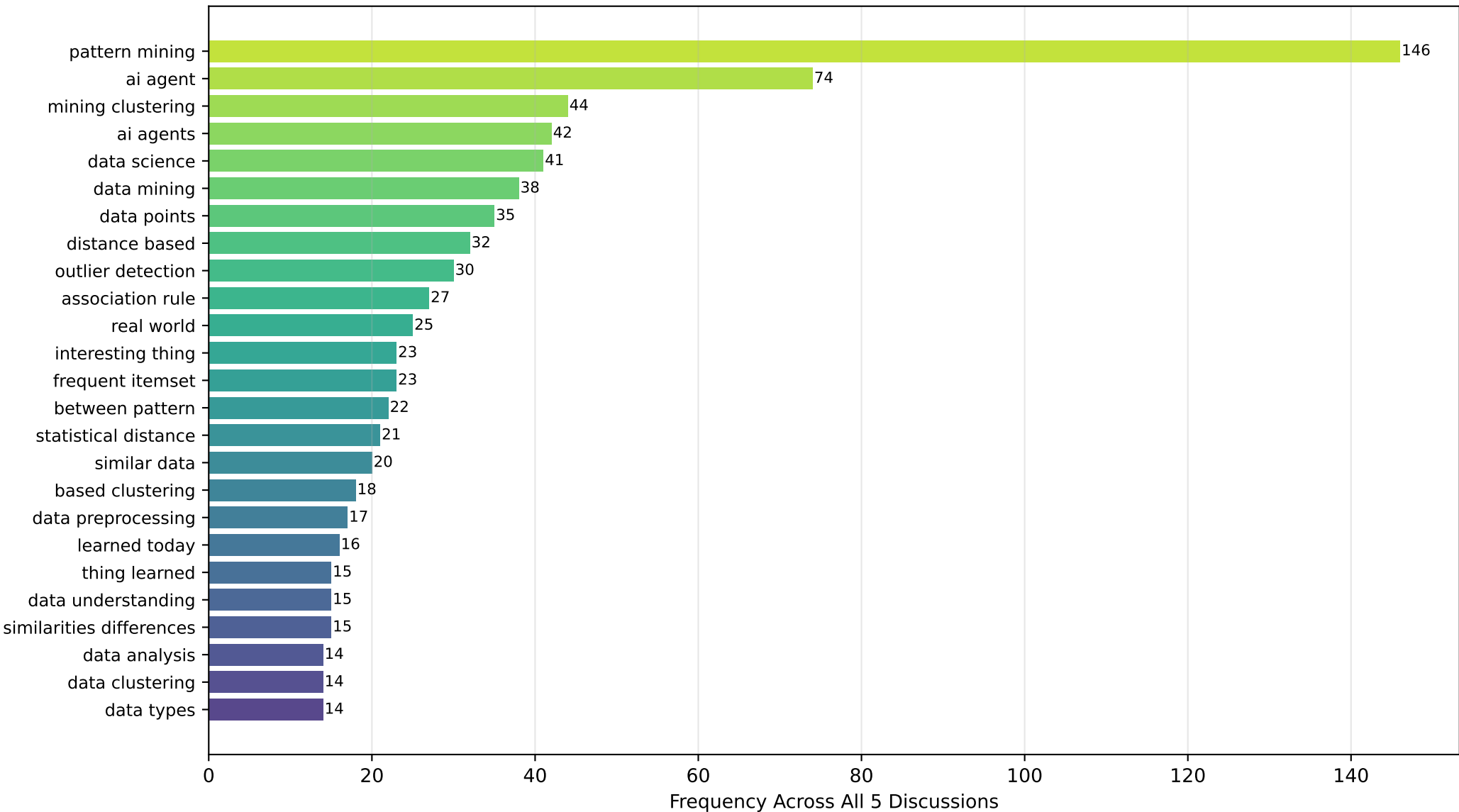
Mean Vocabulary Richness (Gold = Winner)



*Key Finding: D1 wins with the highest mean Type-Token Ratio — students used more diverse words relative to their total word count. Higher TTR = more expressive, less repetitive writing.*

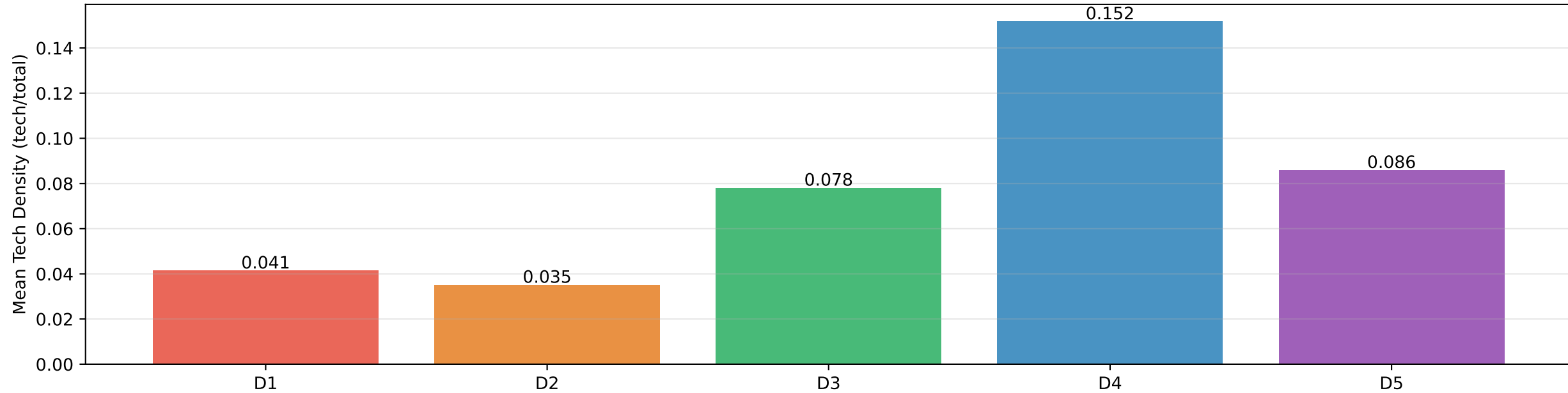
Q14 | What Concept Pairs Co-Occur Most Across All Discussions?

Top 25 Bigrams Across Entire Corpus

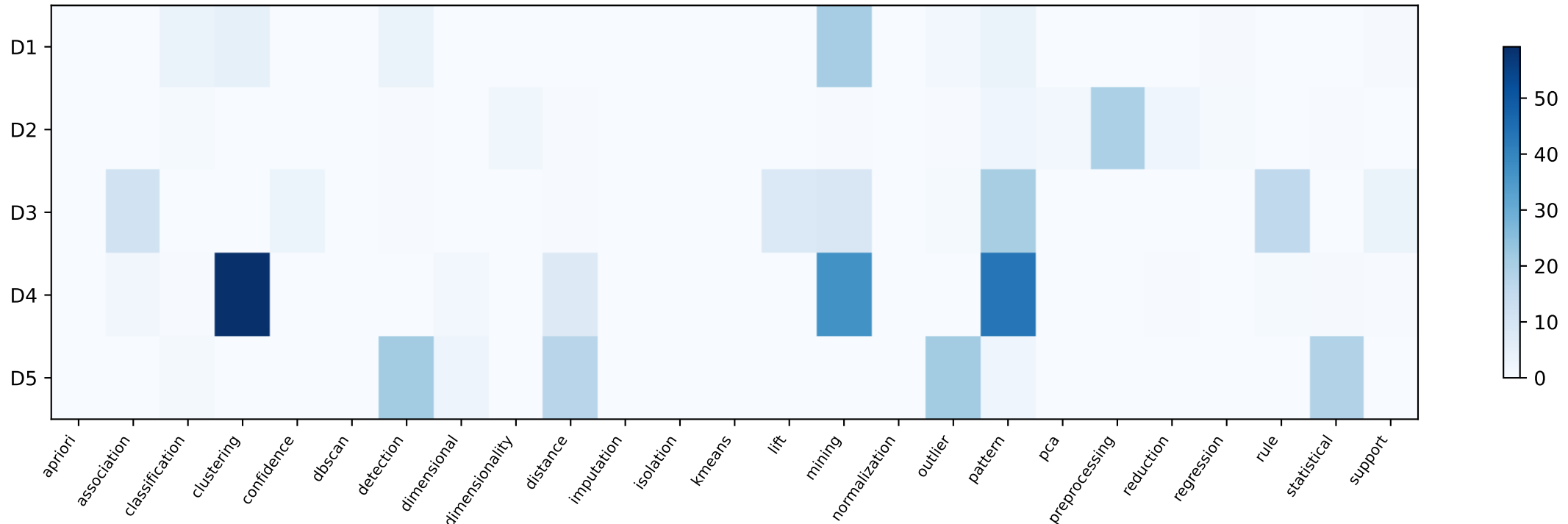


Key Finding: "pattern mining" is the #1 bigram. Compound course concepts dominate: "outlier detection", "distance based", "machine learning", "data science".

Mean Technical Vocabulary Density by Discussion



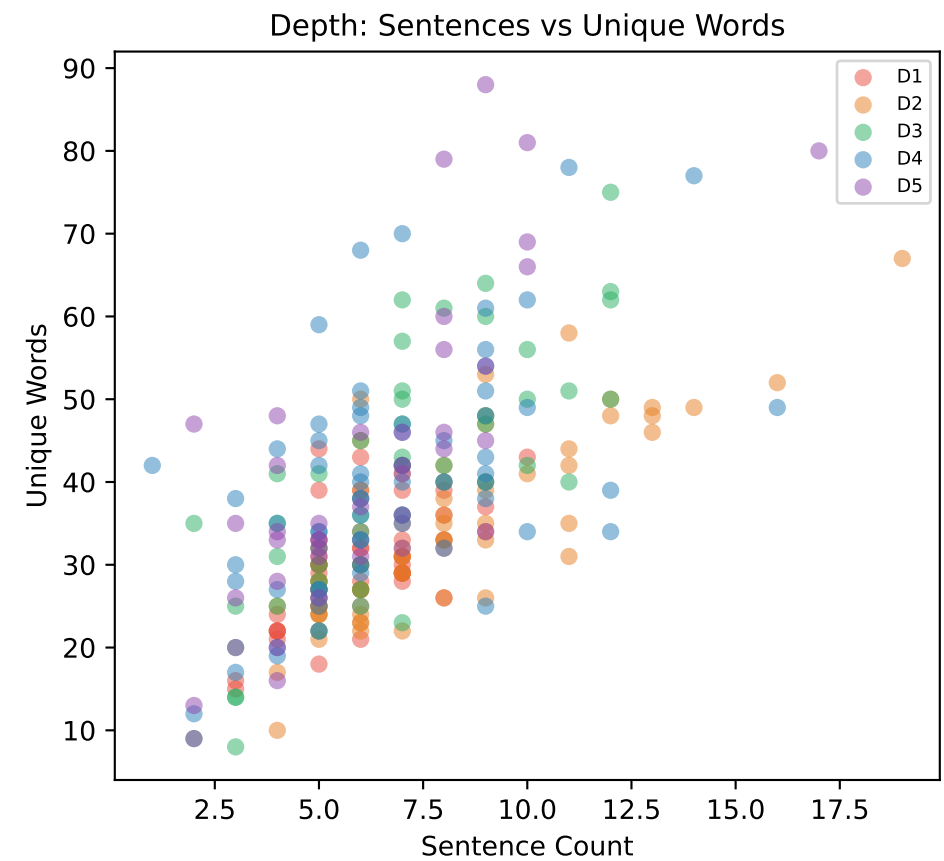
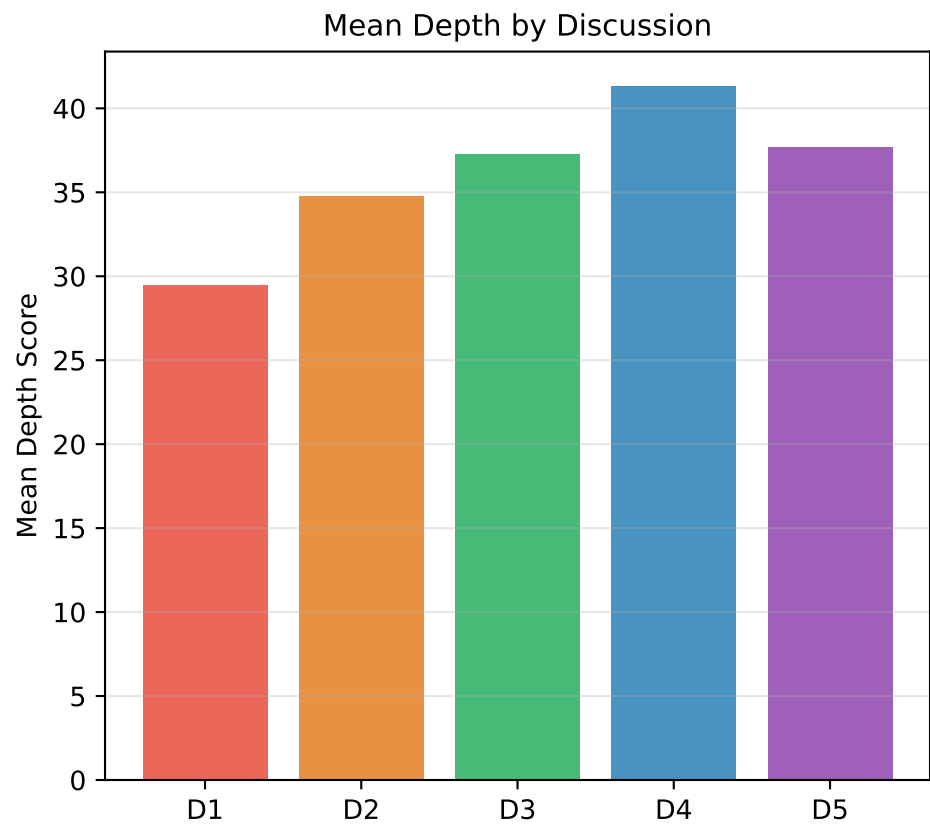
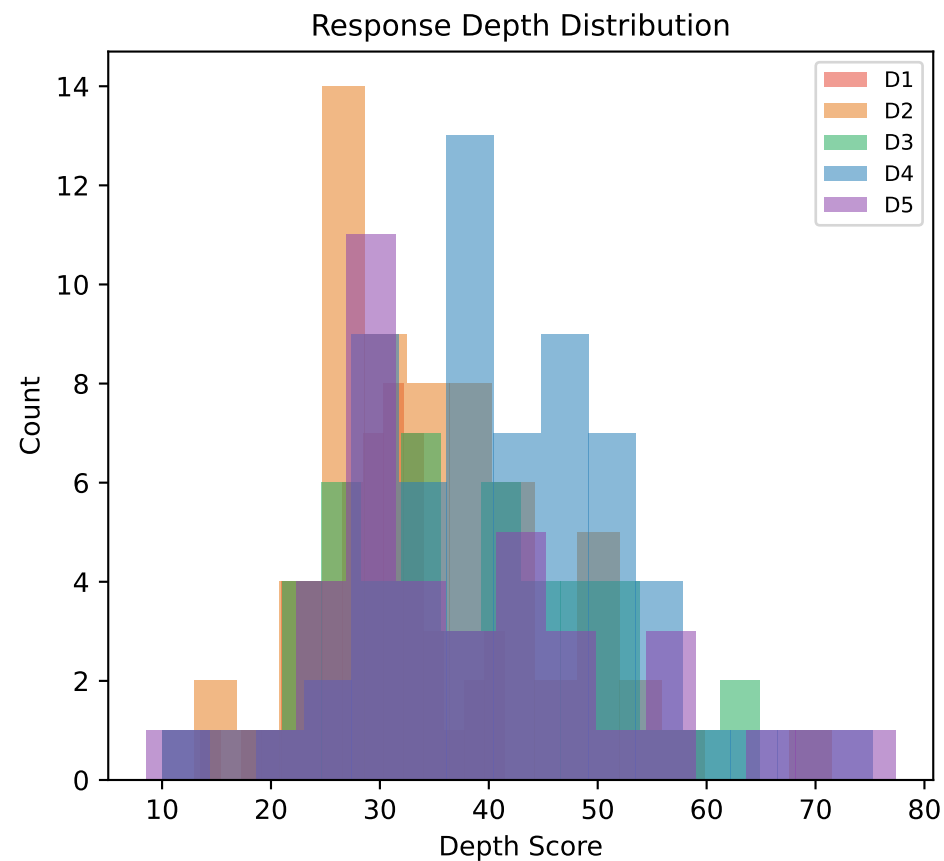
Technical Term Heatmap (mentions per 1000 words)



Key Finding: D5 uses the densest technical vocabulary. D1 is the least technical — students wrote about personal motivations, not algorithms.



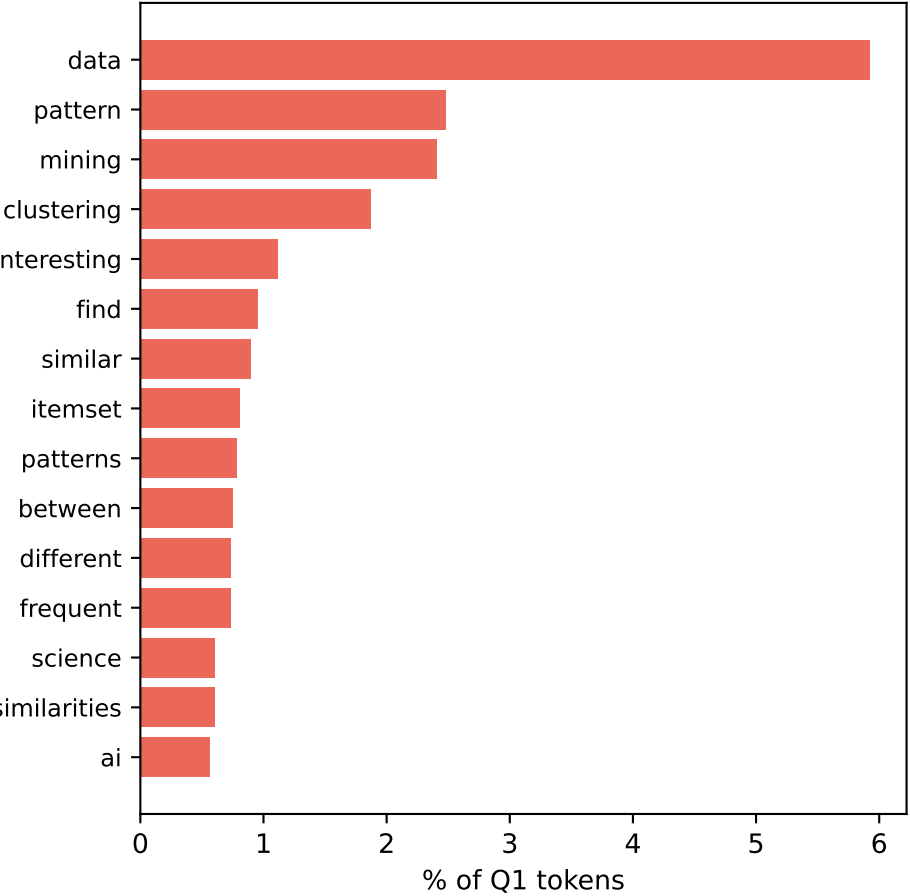
# Q16 | How Deep Are Student Responses? (Composite Depth Score)



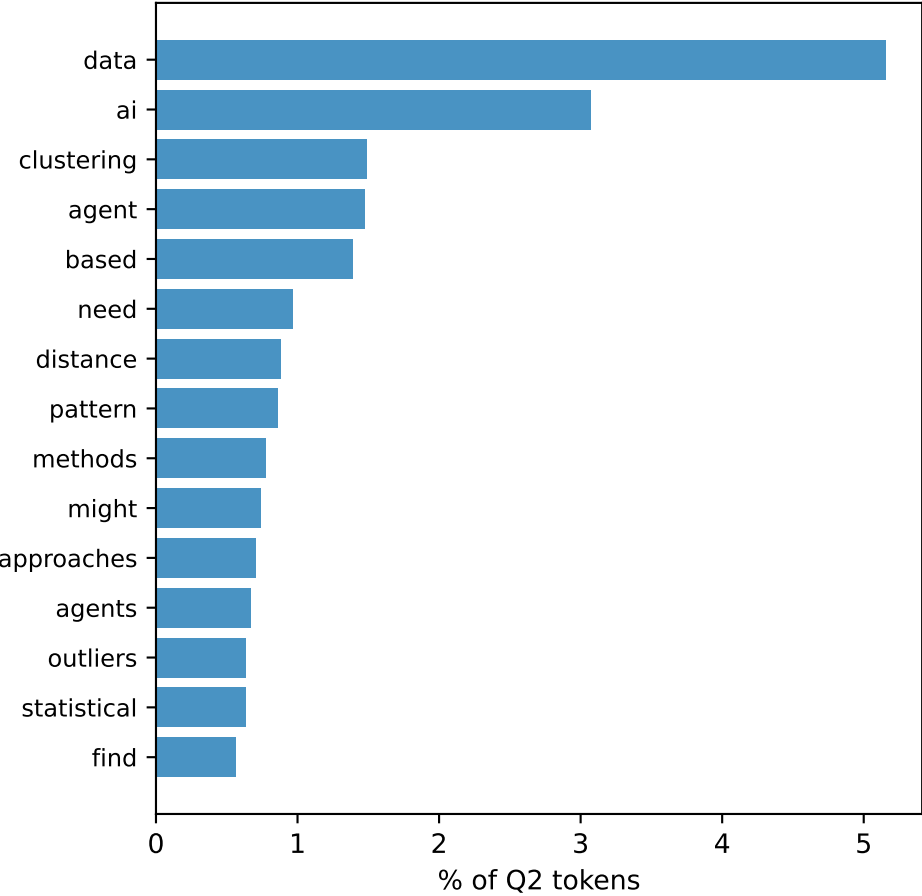
Key Finding: D4 produced the deepest responses on average. Depth = unique words + sentence count + technical density. High within-discussion variance.

Q17 | Do Q1 and Q2 Responses Have Different Personalities?

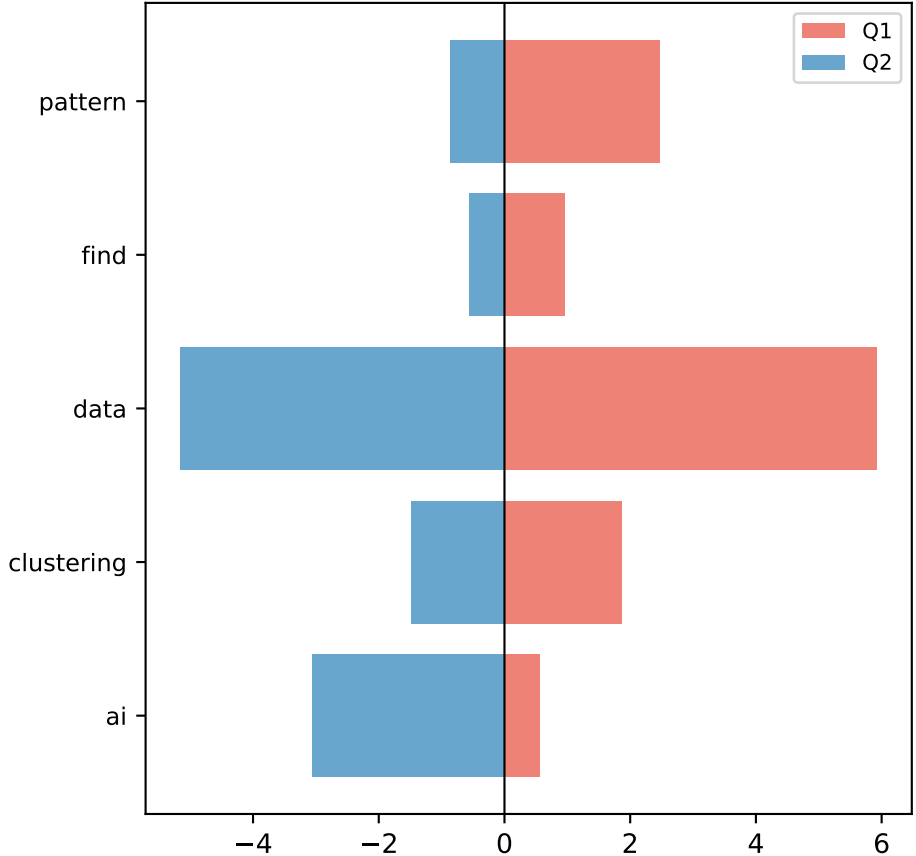
Q1 Top Words



Q2 Top Words



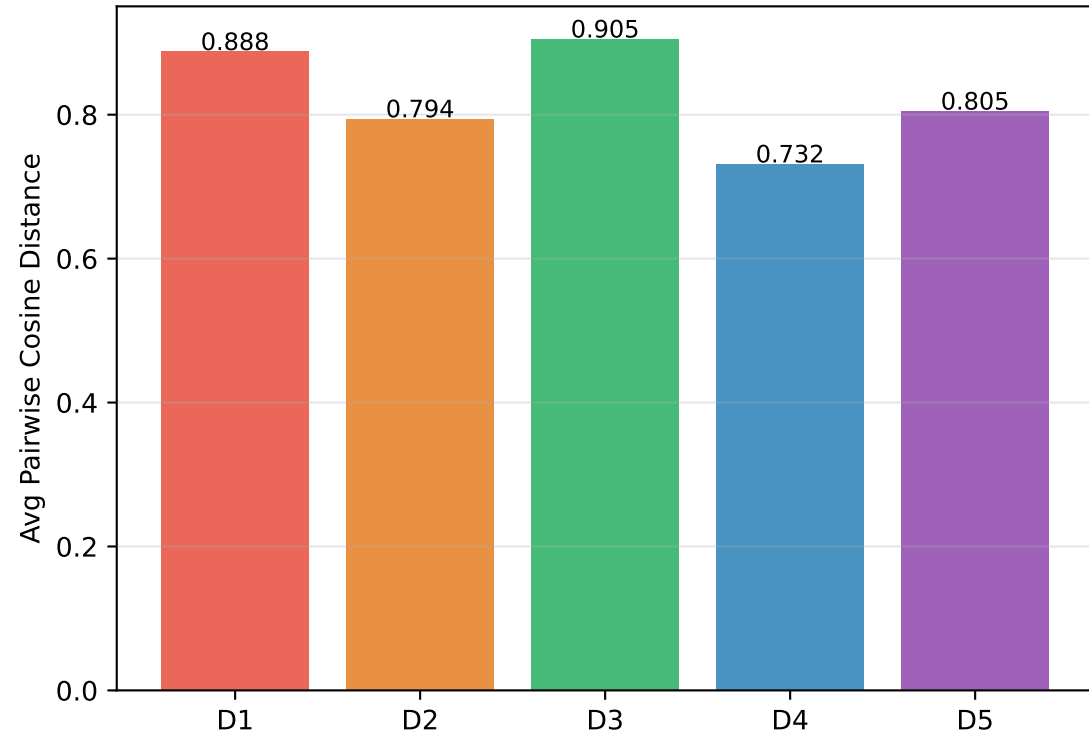
Shared Words: Q1 vs Q2 Freq



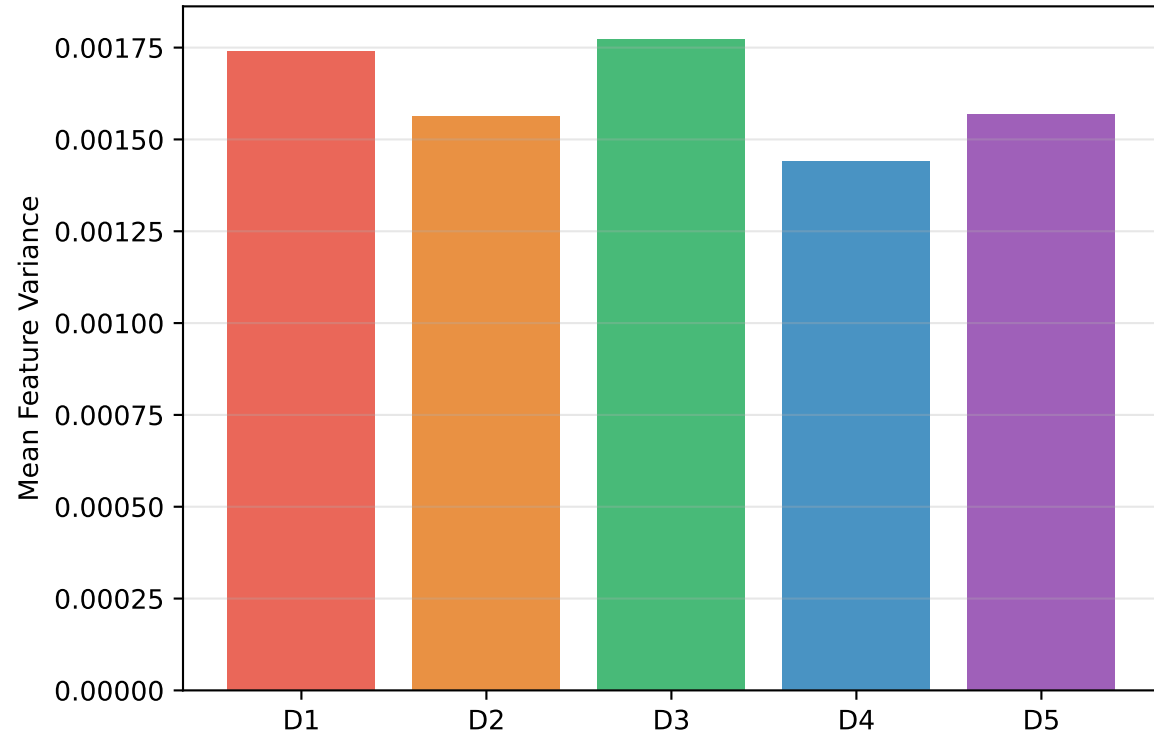
Key Finding: Q1 vocabulary is example-heavy; Q2 is evaluation/comparison-heavy. 5 words overlap in both top-15 lists — the shared conceptual core.

## Q18 | Which Discussion Had the Most Polarizing / Diverse Responses?

Response Diversity (Avg Pairwise Distance)

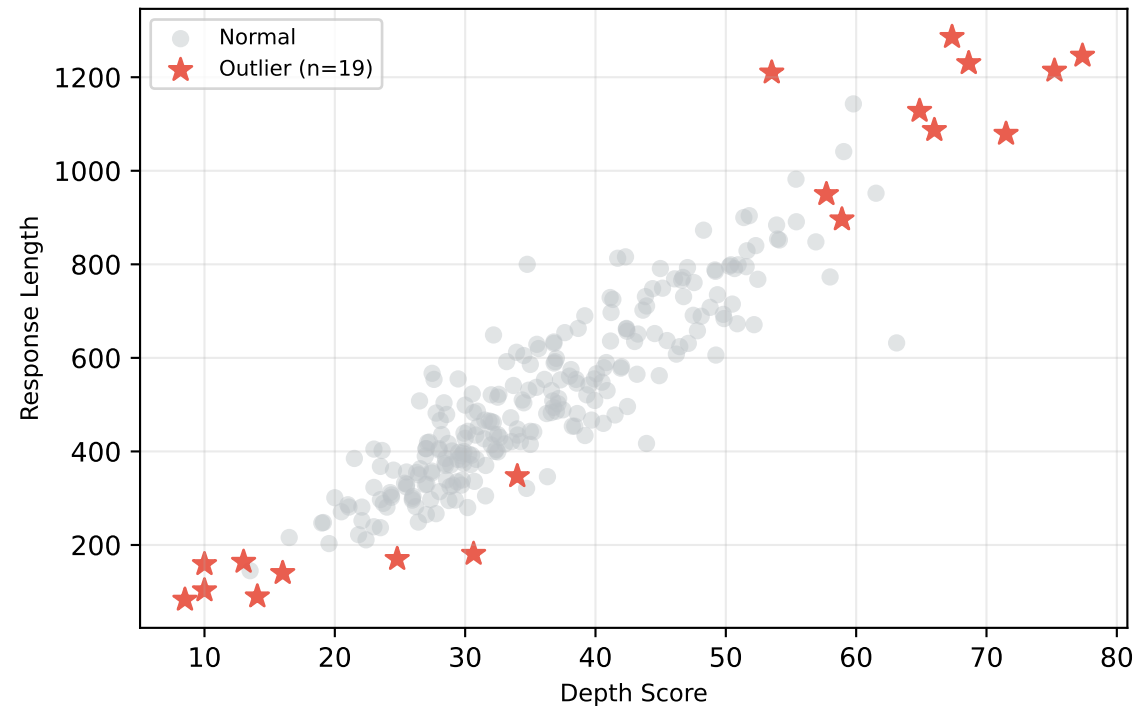
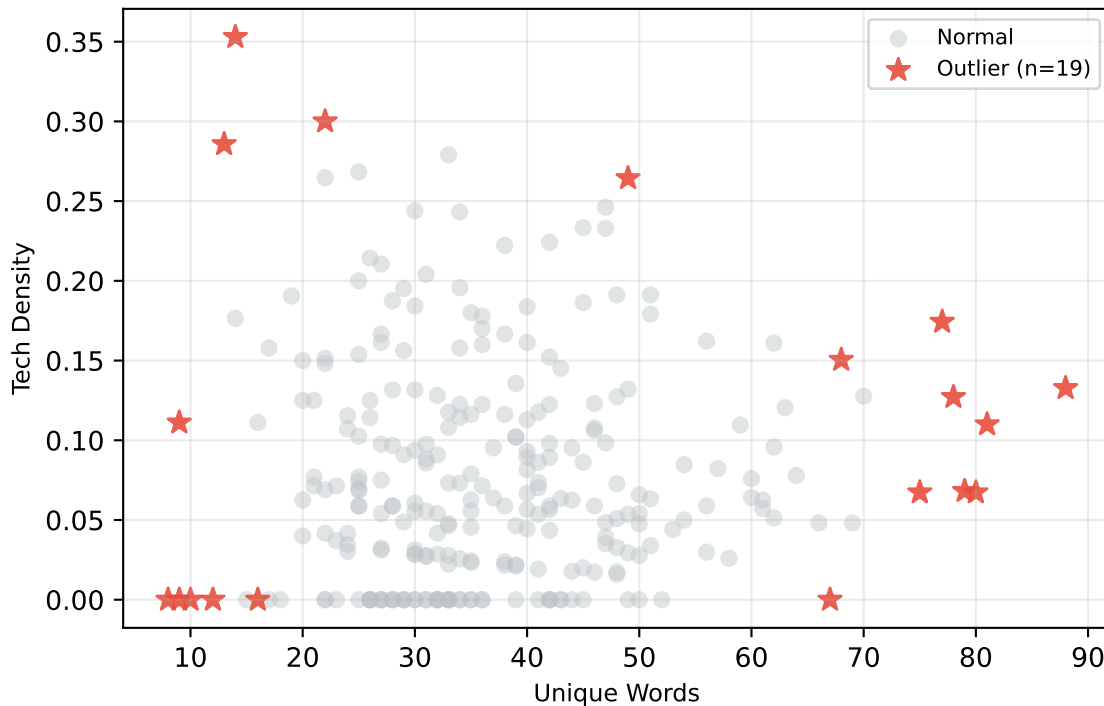
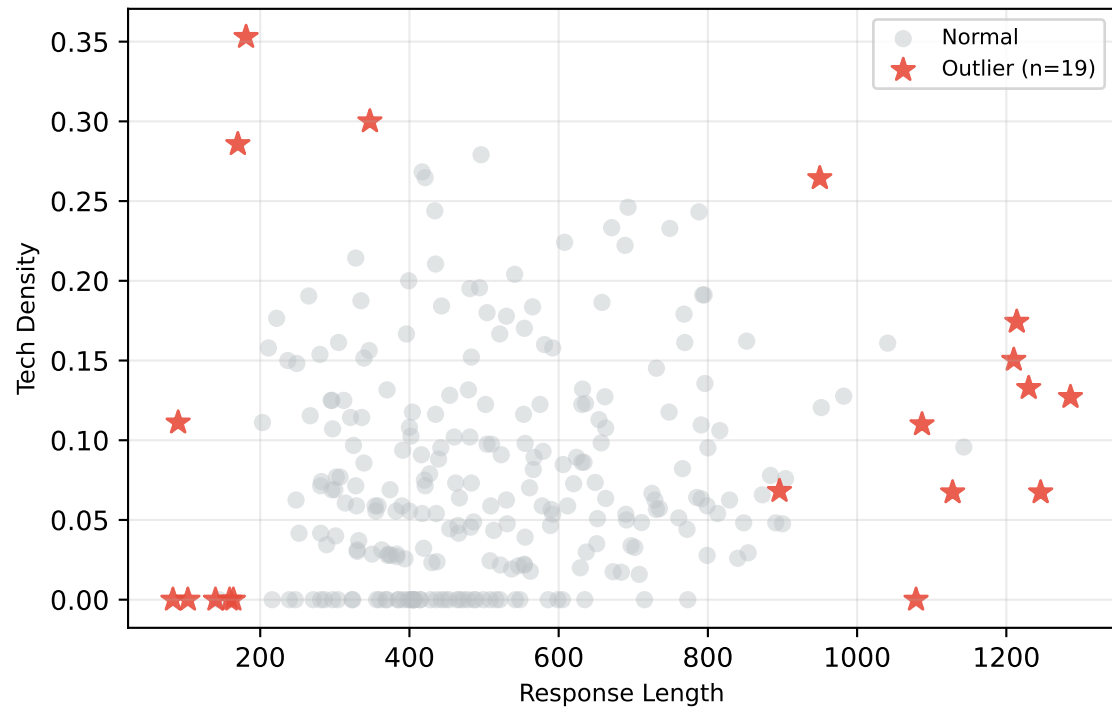
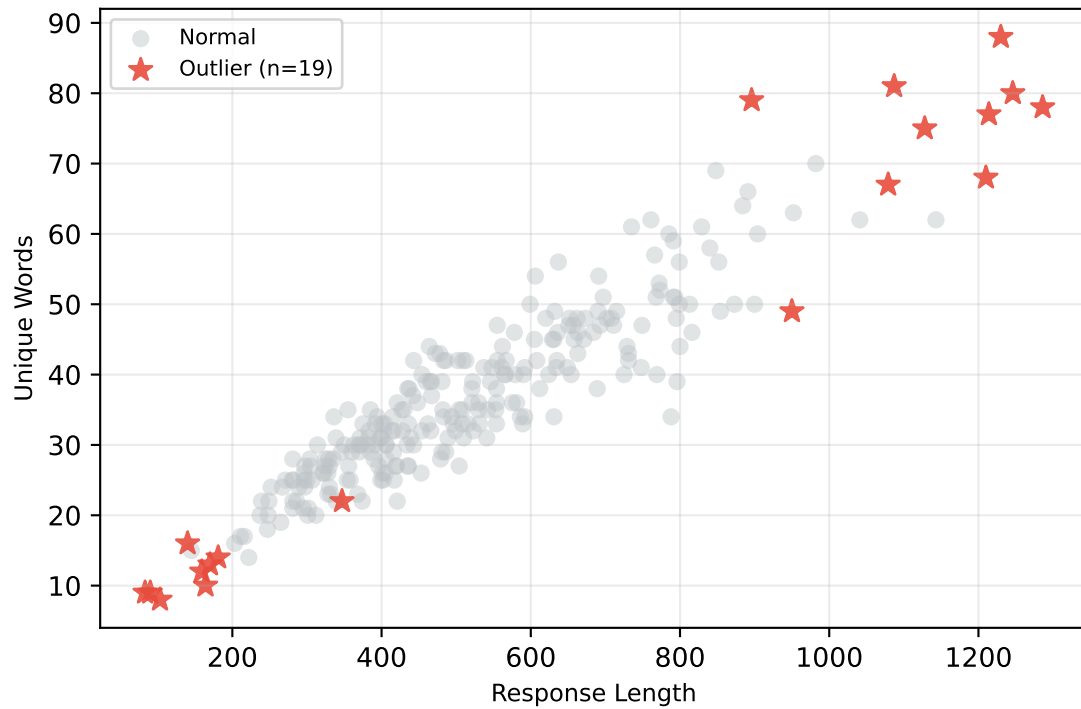


Response Diversity (TF-IDF Variance)



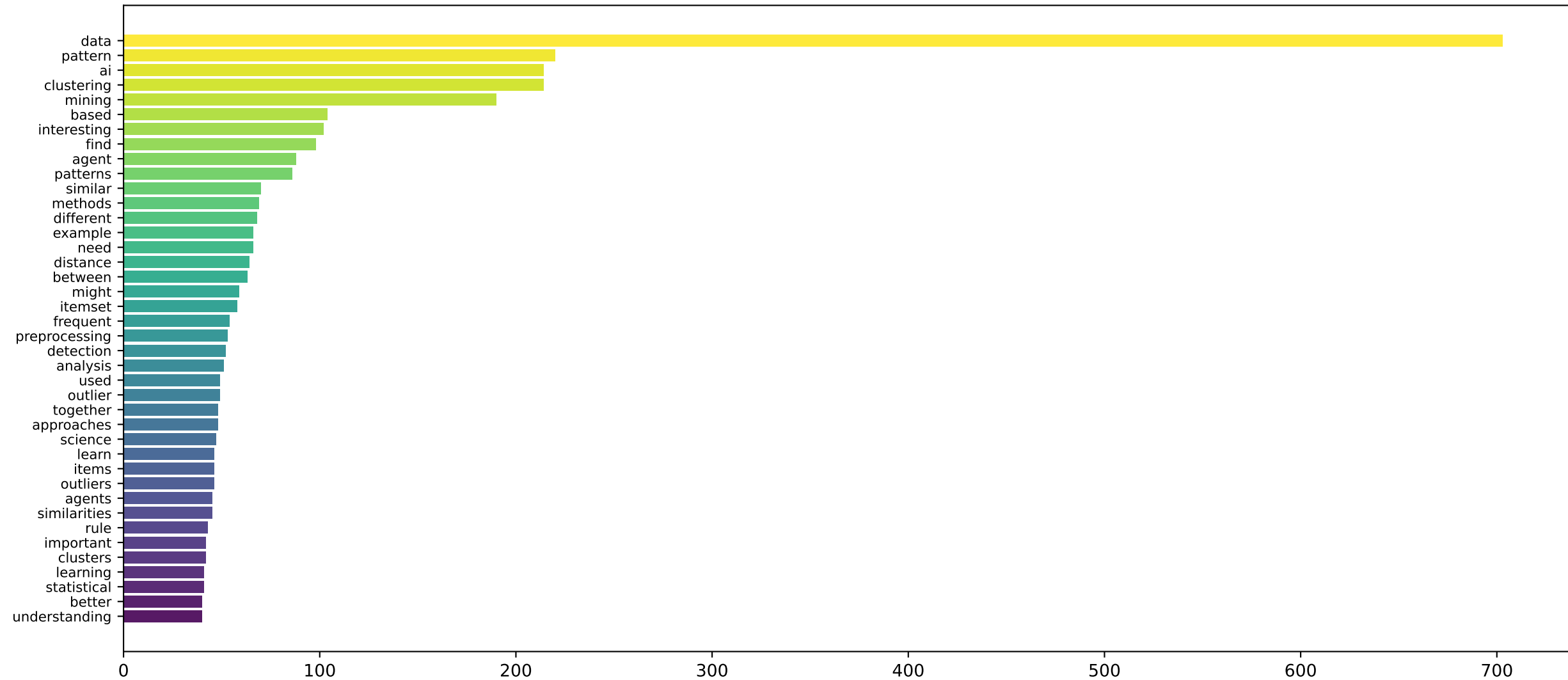
Key Finding: D3 is the most polarizing — responses are furthest apart from each other. D4 is most uniform — students converged on similar answers.

## Q19 | Which Responses Are Outliers Across MULTIPLE Dimensions Simultaneously?



Key Finding: 19 responses are multi-dimensional outliers — unusual in length, vocabulary, AND technical density at once. These are the truly exceptional (or truly minimal) responses.

Top 40 Words Across All Discussions



Key Finding: "Data", "outlier", "detection", "clustering" dominate the full corpus — the course vocabulary has been internalized. D5 cloud shows a striking "AI vs statistical" debate in word form.